



Pediatric Bickerstaff brainstem encephalitis: a systematic review of literature and case series

Jonathan Douglas Santoro¹ · Daniel V. Lazzareschi² · Cynthia Jane Campen¹ · Keith P. Van Haren¹

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Abstract

Objective To characterize the phenotype of pediatric Bickerstaff's brainstem encephalitis (BBE) and evaluate prognostic features in the clinical course, diagnostic studies, and treatment exposures.

Methods We systematically reviewed *PubMed*, *Web of Science*, and *SCOPUS* databases as well as medical records at the Lucile Packard Children's Hospital to identify cases of pediatric BBE. Inclusion required all of the following criteria: age ≤ 20 years, presence of somnolence or alterations in mental status at the time of presentation or developed within 7 days of presentation, ataxia, and ophthalmoplegia.

Results We reviewed 682 manuscripts, identifying a total of 47 pediatric BBE cases. We also describe five previously unreported cases. The phenotype of these pediatric patients was similar to previously published literature. Sixty-eight percent of patients demonstrated positive anti-GQ1b antibody titers, yet the presence of these antibodies was not associated with longer times to recovery. Patients with neuroimaging abnormalities featured a longer median time to recovery, but this was not statistically significant ($p = 0.124$). Overall, patients treated with any form of immunotherapy (intravenous immunoglobulin, steroids, or plasmapheresis) demonstrated shorter median time to resolution of symptoms compared to supportive therapy, although this trend was not statistically significant ($p = 0.277$). Post-hoc t tests revealed a trend towards use of immunotherapy against supportive care alone ($p = 0.174$).

Conclusion Our study identified clinical, radiologic, and treatment features that may hold prognostic value for children with BBE. The role of immunotherapy remains under investigation but may prove of utility with further, randomized controlled studies in this rare disease.

Keywords Pediatric · Bickerstaff · Brainstem · Encephalitis · Immunotherapy

Introduction

Bickerstaff's brainstem encephalitis (BBE) is a rare, immune-mediated, disorder of the brainstem and peripheral nervous system characterized by a clinical triad of ataxia, encephalopathy, and ophthalmoplegia [1, 2]. The

pathophysiology of BBE is poorly understood but is believed to lie on a spectrum with other acute immune-mediated polyneuropathies, such as Guillain-Barré Syndrome (GBS) and Miller-Fisher syndrome (MFS). BBE is unique from these entities, in that it presents with marked alterations in mental status [2, 3]. Many of the existing large-scale studies of BBE have been conducted in Japan where the annual incidence is estimated to be 8 per 100 million individuals, and children constitute a minority of reported cases [2, 4, 5].

To our knowledge, no prior systematic reviews of pediatric BBE exist, nor are there trials validating therapeutic interventions in pediatric populations. This study explores 47 cases of pediatric BBE from an extensive search of medical literature as well as five, previously unreported, cases evaluated at the Lucile Packard Children's Hospital at Stanford University (LPCH). This analysis has three primary objectives: first, to phenotype pediatric BBE, and in

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✉ Jonathan Douglas Santoro
jsantoro@stanford.edu

¹ Division of Child Neurology, Lucile Packard Children's Hospital at Stanford, Palo Alto, CA, USA

² Stanford University School of Medicine, Palo Alto, CA, USA

doing so, identify clinical presentations and neurodiagnostic profiles; second, to identify diagnostic and prognostic biomarkers in pediatric BBE; and third, to evaluate the utility of therapeutic immunomodulatory interventions in children with this disease.

Methods

Following approval by the Stanford University Institutional Review Board, we conducted a literature review to identify appropriate studies and searched the *PubMed* database on April 29, 2016. We ran additional searches in *PubMed*, *Web of Science*, and *Scopus* databases on May 23, 2016, June 13, 2016, and April 24, 2017. No restrictions in date of publication were applied. The authors included manuscripts published in either English or Spanish. The specific search strategies for all databases are included in supplemental materials. Five additional cases were identified through course of practice at LPCH from 2012 to 2017 (Supplemental Data).

Inclusion and exclusion criteria for evaluating the literature were determined prior to the search (“[Appendices A](#)” and “[B](#)”) and were based on previously used criteria published by Koga et al. [4]. All patients had to meet criteria for at least “probable” BBE. All manuscripts were reviewed independently by the authors (JDS and DVL).

Data were extracted from manuscripts with cases identified as meeting all inclusion criteria and no exclusion criteria using a standardized tool for term classification (“[Appendix B](#)”), which was performed in duplicate by the authors (JDS and DVL) to eliminate bias or inconsistencies in data extraction (Fig. 1).

To assess responses to immunotherapy, a one-way analysis of variance (ANOVA) test was performed among four treatment groups: supportive care only, steroid or IVIG monotherapy, steroid/IVIG dual therapy, and any immunotherapy (steroids, intravenous immunoglobulin (IVIg), or plasmapheresis (PLEX)). Additionally, we used a multivariate linear regression to test our primary hypothesis that immunotherapy intervention would reduce the median time to resolution of neurologic symptoms. The exposure was binary and defined as receiving any form of immunotherapy versus receiving supportive and non-immunologic-based therapies only. The outcome was time-to-symptom resolution using propensity score and binary presence/absence of immunotherapy as predictors. To calculate each patient’s propensity score, we used a logistic regression model with multiple binary and continuous predictors, including history of prior upper respiratory tract infection (URTI), severity of altered mental status, bulbar palsy, muscle weakness, and hyperreflexia. For variable selection in this non-parsimonious model, we excluded those measured variables whose coefficients’ *p* values exceeded 0.2 in

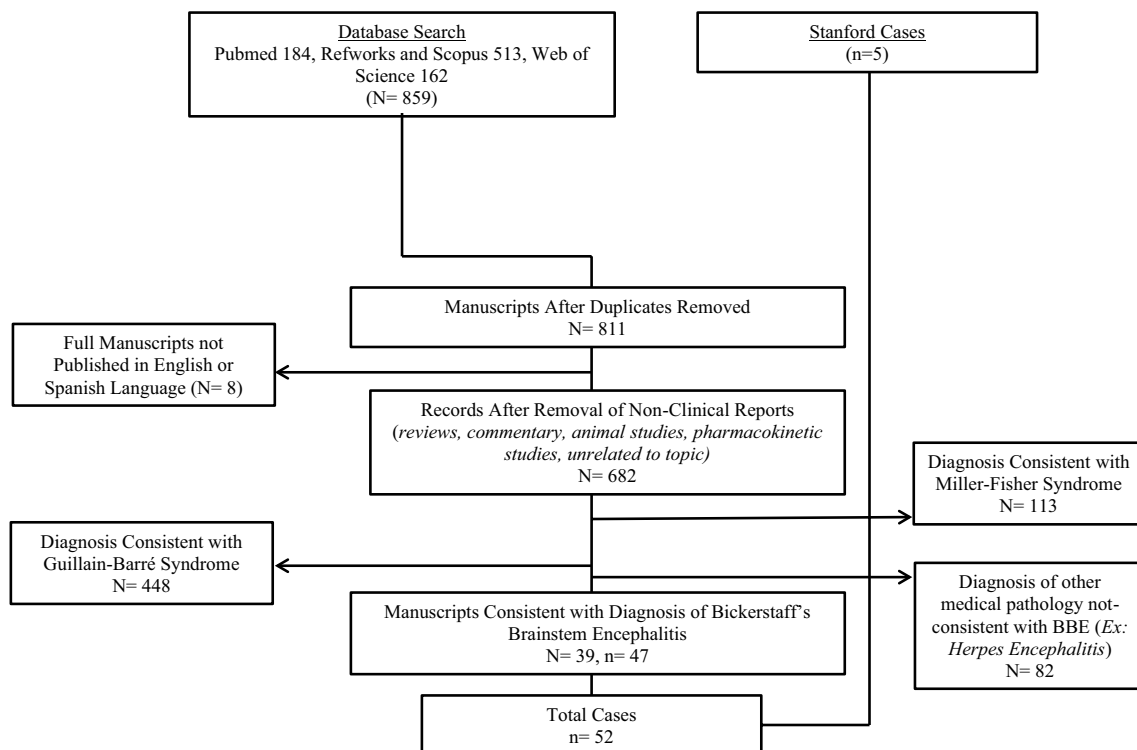


Fig. 1 Manuscript review algorithm (*N* = manuscripts, *n* = cases)

univariate screening. Propensity scores were employed to improve balance between the patients who did and did not receive immunotherapy, and these scores reflect the likelihood of each patient receiving immunotherapy based on the severity of his/her disease status, thereby minimizing severity bias in the evaluation of disease outcome based on treatment. Due to limitations in sample size, score-based matching methods of analysis were not possible, and as such the propensity score was used as a continuous predictor alongside binary immunotherapy exposure in the final linear regression model.

Results

In total, 859 citations were retrieved among three database resources. A combined method of manual and citation manager de-duplication was conducted to eliminate 48 duplicate citations and an additional eight citations that were published in neither English nor Spanish, resulting in a total of 803 citations for review. A review at the abstract level eliminated an additional 121 citations. In total, 682 manuscripts were included for full manual review, with authors (JDS and DVL) determining inclusion based on pre-defined criteria. Following application of strict inclusion/exclusion criteria, this study identified 39 manuscripts describing 47 patients [6–42]. Five additional patients treated at LPCH were also reviewed and included. Of these, there were 32 cases of “definitive” BBE and 20 cases of “probable” BBE [4]. A complete collection of all cases is provided in supplemental materials.

The median age of patients was 10 years; the youngest patient was 2 years of age. Sex was specified in 50 cases, of which 36 were male and 14 were female (2.6:1 male-to-female ratio). A total of 47 patients (90%) reported a transient, self-resolving, illness preceding the onset of neurologic symptoms with a median time of antecedent illness to neurologic symptom development of 9 days. Of these patients, 14 (30%) experienced a flu-like illness; 19 (40%) reported an upper respiratory tract infection (URTI); 13 (25%) reported gastroenteritis; and 5 (10%) reported experiencing both a flu-like illness or URTI and gastroenteritis.

Beyond the characteristic triad of altered mental status, ataxia, and ophthalmoplegia, the most commonly reported neurologic findings in the 52 cases were hypo/areflexia in 33 patients (63%), extremity weakness in 29 (56%), bulbar palsy in 25 (48%), visual deficits/disturbances including diplopia in 22 (42%), facial palsy in 20 (38%), hyperreflexia in 13 (25%), and altered sensorium or paresthesias in 13 (25%) (Table 1). Intubation for mechanical ventilation, a post hoc marker for severity of disease, was required in 9 patients (17%). Of this group, 3 (33%) had positive MRI

Table 1 Preceding illness, therapeutic intervention and neurodiagnostic studies by frequency and time to improvement

	Frequency <i>n</i> (%)
Demographics	
Median age (years)	10
Gender (M:F)	36:14
Preceding illness (<i>n</i> = 47)	
Any illness	42 (89)
URTI	19 (40)
Flu-like syndrome	14 (30)
Gastroenteritis	13 (28)
None	5 (10)
Clinical presentation^a (<i>n</i> = 52)	
Hypo/areflexia	33 (63)
Extremity weakness	29 (56)
Bulbar palsy	25 (48)
Visual deficits	22 (42)
Facial palsy	20 (38)
Hyperreflexia	13 (25)
Altered sensorium/paresthesias	13 (25)
Babinski reflex	10 (19)
Nystagmus	8 (15)
Hypotonia	8 (15)
Neck stiffness	7 (13)
Hypertonia/rigidity	2 (4)
Aphasia	2 (4)
Clinical course (<i>n</i> = 44)	
Median days to nadir ^b	10
Median days to full resolution	90
Treatment (<i>n</i> = 49)	
Supportive	15 (31)
Steroids and IVIg	12 (24)
Acyclovir	11 (22)
IVIg only	9 (18)
Antibiotics	9 (18)
Steroids	7 (13)
IVIg and plasmapheresis	2 (4)
Plasmapheresis	1 (2)
Neurodiagnostic studies	
Abnormal EEG	24/33 (72)
Anti-GQ1b Ab positivity	19/28 (68)
CSF pleocytosis	12/31 (32)
CSF cytoalbuminologic dissociation	10/31 (32)
Abnormal CT	4/10 (40)
Abnormal MRI	13/32 (41)
Abnormal EMG/NCS	12/27 (44)

^aIn addition to altered mental status, ophthalmoplegia, and ataxia

^bAlternatively defined as days to first clinical improvement

findings in the brainstem and 3 (33%) were GQ1b positive. The median time of intubation was 3 days.

Neurodiagnostic testing revealed 19 of 28 (68%) patients with anti-GQ1b antibody positivity, 12 of 27 (44%) with abnormal EMG or nerve conduction studies (NCS), 13 of 32 (41%) with non-specific abnormalities on brain MRI (predominantly T2 white matter hyperintensities), and 24 of 33 (72%) with non-epileptic abnormalities in electroencephalographic tracings (predominantly diffuse slowing and excessive delta activity). Patients with MRI abnormalities had the longest median time to symptom resolution at 120 days and demonstrated a median time to clinical nadir at 9 days. The former metric (median time to symptom resolution) was statistically insignificant when compared to patients without neuroimaging abnormalities ($p = 0.124$). EMG/NCS data were too limited to evaluate in association with outcome.

All 52 patients in this study had a lumbar puncture (LP). A total of 31 (60%) patients had an LP, which demonstrated cerebrospinal fluid (CSF) abnormalities. Of these 31 patients, 24 (77%) occurred within the first week of hospitalization, while the remaining seven patients (23%) were discovered on serial lumbar punctures following previously normal findings. Of the 31 patients with available CSF studies, 17 (55%) showed elevated protein; 12 (39%) demonstrated pleocytosis; 10 (32%) demonstrated albuminocytologic dissociation; and 3 (10%) showed decreased glucose (Table 1). The percentage of CSF profiles with one or more abnormality within one week of symptom onset was 61.5%, increasing by 10.7% with additional samples obtained after the week of hospital admission.

During hospitalization, 15 (31%) patients received supportive therapy; 12 (24%) received steroid–IVIg dual therapy; 9 (18%) received IVIg monotherapy; 7 (13%) received steroid monotherapy; 2 (4%) received IVIg with PLEX; and 1 (2%) received PLEX only. Additionally, 11 (22%) of patients were exposed to acyclovir and 9 (18%) were exposed to antibiotics in the first week of hospitalization. The median time to resolution of symptoms in those treated with steroid monotherapy, IVIg monotherapy, supportive therapy, IVIg and steroid dual therapy, PLEX, and PLEX–IVIg dual therapy was 24, 60, 73, 102, 120, and 171 days, respectively. Of the 19 patients with GQ1b Ab positivity, all but one received immunotherapy, with a median time to nadir of 10 days and a median time to full recovery of 111 days.

A one-way ANOVA comparing supportive care, steroids and IVIg dual therapy, steroids or IVIg monotherapy, and any immunotherapy was statistically insignificant ($p = 0.277$). Sub analysis with Tukey's post hoc comparing steroids and IVIg dual therapy, steroids or IVIg monotherapy, and any form of immunotherapy against supportive therapy revealed p values of 0.757, 0.287, and 0.174, respectively (Table 2 and Fig. 2). We used propensity scores to account for illness severity while comparing the use of any

immunotherapy versus supportive and/or non-immunologic (e.g., antibiotics) therapies on symptom mitigation but found no significant predictive value for immunotherapy regarding time to nadir or full resolution ($p = 0.941$ and $p = 0.370$, respectively). There were no statistically significant differences between “probable” and “definitive” classifications of BBE with regards to use of immunotherapy [4].

Discussion

This study provides the first comprehensive case analysis of BBE disease course, diagnostic studies, response to immunotherapy, and prognosis in pediatric populations. Pediatric BBE has a similar clinical profile to adults, although neurodiagnostic investigations with LP, MRI, EMG/NCS and EEG appear to be more routinely abnormal in the pediatric population. Despite statistical insignificance, the trending p values for use of immunotherapy in our varied cohort may suggest a role for immunomodulatory intervention in this population based on earlier median times to clinical nadir and total resolution of symptoms in those treated with steroids, IVIg, or a combination of these therapies.

The clinical presentations of pediatric BBE in our cohort appeared similar to previously published adult cases, albeit with a higher prevalence in males [2, 4]. Biomarkers were analyzed for frequency of abnormalities in this study. Our study found a larger percentage of abnormal MRI studies compared to previous investigations of pediatric BBE (41 vs. 14%) [5]. Those with abnormal neuroimaging on MRI had longer median time to full recovery in our cohort although this was not clinically significant. Of the pediatric patients with reported autoimmune markers, 68% of pediatric patients demonstrated positive anti-GQ1b antibody titers, which is comparable to adult studies (69%) [2, 4, 43, 44]. The underlying etiology of antibody production to the GQ1b moiety remains unclear, and the diagnostic relevance of assessing these titers merits further investigation. Prior studies suggest a lower prevalence of anti-GQ1b antibodies in children (43%), but these serum samples were obtained within 3 months of symptom onset, and given the unknown duration of auto-antibody presence in circulation, detectable titers may have lapsed by the time of sampling, resulting in falsely low positivity [5]. Ultimately, CSF profiling in our study appeared to be consistent with previously reported literature. With the exception of one pediatric case, anti-GQ1b antibody negativity did not co-occur with marked CSF pleocytosis as has frequently been observed in adult cases [4]. This study further demonstrated that the percentage of abnormal CSF profiles within 1 week of symptom onset was comparable in children and adults at 61.5 and 56%, respectively, although the percentage of adults with abnormal CSF surveyed 2 weeks from symptom onset increased by 26%

Table 2 Demographics and clinical profiles by therapeutic interventions received

	Supportive therapy ^a (<i>n</i> = 15)	IVIg and steroids (<i>n</i> = 12)	IVIg or steroid monotherapy (<i>n</i> = 18)	Any immunotherapy ^b (<i>n</i> = 31)
Demographics				
Median year of publication	1987	2009	2005	2006
Age (mean, IQR) ^c	9.8 years [7.5]	8.8 years [9.2]	9.9 years [7.5]	9.4 years [8.5]
Gender (M:F)	7:7	10:2	14:3	25:6
Preceding symptoms: <i>n</i> (%)				
URI	4/14 (29)	3/12 (25)	11/18 (61)	14/31 (45)
Flu-like syndrome	4/14 (29)	3/12 (25)	4/18 (22)	8/31 (26)
Gastroenteritis	3/14 (21)	5/12 (42)	2/18 (11)	7/31 (23)
None	1/14 (7)	1/12 (8)	3/18 (17)	4/31 (13)
Days prior to presentation (mean)	20.9 days	10.7 days	9 days	9.7 days
Clinical presentation: <i>n</i> (%)				
Hypo/areflexia	10/15 (67)	5/12 (42)	14/18 (78)	20/31 (65)
Extremity weakness	7/15 (47)	8/12 (67)	10/18 (56)	19/31 (61)
Bulbar palsy	6/15 (40)	8/12 (67)	9/18 (50)	18/31 (58)
Facial palsy	8/15 (53)	5/12 (42)	7/18 (39)	12/31 (39)
Hyperreflexia	3/15 (20)	6/12 (50)	4/18 (22)	10/31 (32)
Intubation	2/15 (13)	3/12 (25)	4/18 (22)	7/31 (23)
Neurodiagnostic studies, <i>n</i> (%)				
Abnormal EEG	6/10 (60)	8/9 (89)	7/11 (64)	16/21 (76)
Anti-GQ1b Ab positivity	0/0 (0)	5/10 (50)	9/12 (75)	15/23 (65)
Abnormal CSF	9/15 (60)	10/12 (83)	13/18 (72)	24/31 (77)
Abnormal CT	4/7 (57)	0/0 (0)	0/2 (0)	0/2 (0)
Abnormal MRI	3/3 (100)	6/11 (55)	3/15 (20)	10/27 (37)
Abnormal EMG	3/7 (43)	2/8 (25)	6/8 (75)	8/17 (47)
Recovery (mean, median, IQR) ^c				
Days to clinical nadir	14.9, 9.5 [12.3]	18.5, 14 [7.5]	13.4, 7.5 [6.75]	15.5, 10 [8.5]
Days to full resolution	212.1, 73 [282.3]	124.7, 102 [118]	91.4, 60 [100]	107.2, 90 [96]
Significance ^d vs. supportive	–	<i>p</i> = 0.757	<i>p</i> = 0.287	<i>p</i> = 0.174

^aSupportive therapy includes antibiotics or antiviral medications but no immunotherapy^bAny immunotherapy includes steroid monotherapy, IVIg monotherapy, steroid and IVIg dual therapy, or any combination with PLEX^cInterquartile range is between 25th and 75th percentiles^dPost hoc *t* test *p* value

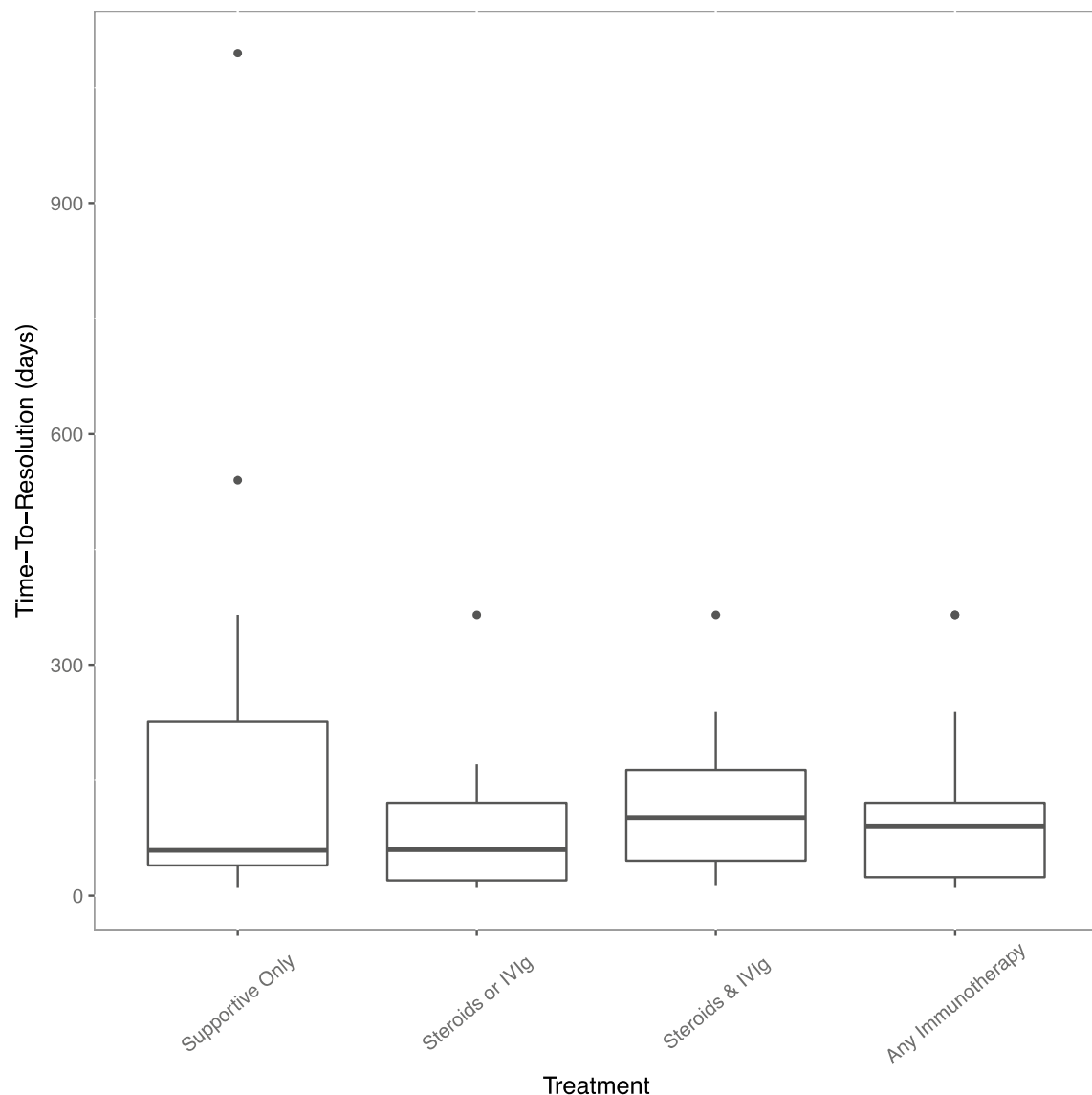


Fig. 2 Intervention type by time to resolution of symptoms (days)

compared to only 10.7% of children. While the immunologic mechanisms underlying this discrepancy remain unclear, these data demonstrate that serial lumbar punctures may be less diagnostically useful in pediatric patients. Neurodiagnostic studies assessed in this cohort lack sensitivity and specificity in this disease process and thus require more nuanced scrutiny before becoming reliable components of prognostic evaluation.

Due to lack of existing treatment algorithms in pediatric BBE, therapeutic interventions and responses were scrutinized closely. Interestingly, patients treated with steroid and IVIg monotherapies demonstrated shorter median times to complete resolution of symptoms at 24 and 60 days, respectively, while other treatment exposures were associated with longer symptom resolution

windows. Time to clinical nadir did not appear to correlate with any specific therapy in our study. Ultimately, more pediatric patients were treated with supportive therapy alone compared to adult cases (31 versus 11%) and among this group there was a median time to recovery of 73 days [40]. Severity bias may explain the shortened median time to recovery of patients receiving supportive therapy only compared to more aggressive dual immunotherapy with IVIg and steroids (102 days), as the median time to recovery was over 40 days longer than either individual immuno-monotherapy. Only three patients were treated with PLEX in our study: 2 in combination with IVIg, one of which was complicated by anaphylaxis after IVIg. Although both steroid and IVIg monotherapies reduced time to resolution of symptoms, available data

are too limited to draw definitive conclusions about treatment outcome. Further, the longer time to full recovery of patients treated with IVIg and steroids (102 days), PLEX alone (120 days), and PLEX with IVIg (171 days) may be similarly skewed by severity bias. Together, these findings suggest a possible response to immunotherapy in pediatric BBE, although clear associations between diagnostic criteria, treatment algorithms, and long-term outcomes are difficult to ascertain, as previously demonstrated in an adult-based Cochrane Review in 2007 [45].

Direct statistical comparison of treatment modalities was attempted but ultimately underpowered, and as such, not significant. It appears that there may be a role for immunotherapy in BBE (steroids and/or IVIg) as it was noted to be safe (only one reported episode of anaphylaxis in response to IVIg and no deaths reported in any study) and non-inferior to supportive therapy with reduced median duration of symptoms. The substantial heterogeneity in treatment of BBE further complicates the underpowered statistical analysis of our sample. Use of these agents in BBE differs from the primary treatment of GBS, a related entity for which steroids have been deemed inefficacious. Similarly, PLEX was infrequently used in patients with BBE in contrast to other pediatric neuro-immune disorders such as GBS, MFS, and myasthenia gravis, and these treatment discrepancies may underscore a lack of understanding of the presumed autoimmune etiology of BBE.

This study had several limitations, most notably the retrospective nature of the identification and clinical course of included patients, as well as the incomplete outcome data available from a variety of case studies. Additionally, data collected in this review were unable to be statistically compared to those of prior studies due to the absence of raw data published in these studies' appendices [2, 4, 43, 44]. The rarity of BBE makes prospective studies difficult, especially in pediatric patients, and thus existing literature outside of this manuscript has been scarce. For this reason, a severity bias, as addressed above, and publication bias in terms of the literature and cases available for review is likely present in our data. Additionally, studies utilizing supportive therapy alone were published earlier (median year of publication was 1987) compared to those with patients who received any immunotherapy (2006), which adds availability heuristic-bias complicating the interpretation of earlier studies whose authors may have utilized immunotherapy if it had been more readily available or more commonly utilized at that time. Similarly the country of origin of reports may have also limited use of particular immunotherapy for access reasons. An additional limitation was that only 54% of the included patients were tested for anti-GQ1b antibodies, 23% reported incomplete outcome data including time to clinical nadir and/or full resolution, and 6% failed to specify any treatment course.

It should also be noted that none of the reference population studies used for adult comparison excluded children, and a small minority of pediatric cases may have skewed the results closer to those observed in our exclusively pediatric assessment, thereby rendering conservative estimates of differences between these populations. Finally, given the rarity of BBE, a small number of patients meeting strict inclusion/exclusion criteria further limits the overall power of our observations and prevents extrapolation of these data to directly inform clinical guidelines. However, this rigidity was needed to rule out less convincing cases which ranged on the brainstem/rhombencephalitis spectrum although we could not rule out overlap between BBE and other pathology in this study.

This study included a systematic review of published cases of pediatric-onset BBE, a unique and under-investigated subset of the disease, as well as previously unpublished pediatric cases from LPCH. Clinical presentation between age groups appeared similar, but variation in yield of neurodiagnostic studies and their potential prognostic value warrants further investigation. Furthermore, use of immunotherapy with either IVIg or steroids may reduce total duration of symptoms, although it is difficult to determine this definitively, given both severity and publication bias. This study highlights the clinical criteria and diagnostic studies needed to thoroughly evaluate pediatric patients for BBE and provides initial data to support the potential use of immunotherapy in these patients. Ultimately, these and similar data exist on a convoluted spectrum of immune-mediated neurological disorders, and rigorous analysis in larger, prospective studies is needed to evaluate the diagnostic and therapeutic utility of these observations.

Compliance with ethical standards

Conflicts of interest All authors declared that they have no potential conflict of interest.

Ethical standards For this type of study formal consent is not required.

Appendix A: Patient related inclusion and exclusion criteria for cases

Inclusion criteria (must meet all criteria)	Exclusion criteria (excluded if any criteria met)
	Study did not meet all inclusion criteria

Inclusion criteria (must meet all criteria)	Exclusion criteria (excluded if any criteria met)
Progression to complete ophthalmoplegia (or alternatively partial ophthalmoplegia at presentation)	Comorbidities or prior medical history that would alter assessment of ophthalmoplegia
Evidence of somnolence or alterations in mental status at the time of presentation or developed within 7 days of presentation without other identifiable cause	Comorbidities or prior medical history that would alter assessment of mental status
Ataxia	Comorbidities that would alter evaluation of ataxia
≤ 20 years of age	≥ 20 years of age
	Comorbidities or prior medical history that would alter immune function
	Presence of sepsis or viremia in a patient
	Seizures at or before presentation
	Prior or concurrent diagnosis of viral or bacterial meningitis or encephalitis
	Prior neurosurgical interventions
	Case had insufficient details to determine if all inclusion criteria were met

Non-patient related exclusion criteria: review articles without case reporting, commentary or editorial pieces, studies in non-human species, pharmacokinetic or genetic studies not relevant to the clinical syndrome in question, and manuscripts that were non-clinical in nature

Appendix B: Standardized manuscript and case review criteria

Antecedent illness: reported as described in manuscripts unless not listed. For terms not listed, the following terminology were used to determine antecedent illness:

URTI: cough, cold, nasal discharge, increased work of breathing, fever.

Flu-like syndrome: fever, headache, myalgia, nausea, emesis, and/or general malaise.

Gastroenteritis: diarrhea (non-bloody), abdominal pain, abdominal cramping, loose stools, emesis, fever.

Symptom classification: clinical presentation of cases was determined to include ataxia, somnolence or altered mental status and ophthalmoplegia based on use of any of the included terms:

- Ataxia: truncal/appendicular, gait instability, wide-based gait, dysmetria, dysdiadochokinesia, cerebellar ataxia, postural instability, positive Romberg.
- Somnolence: any form of altered mental status (irritability, confusion, somnolence, drowsiness, fluctuating levels of consciousness, excessive sleeping, coma, etc.).
- Ophthalmoplegia: any limitation of ocular movement, ptosis, nystagmus, any pupillary defect (sluggish pupillary reflexes, anisocoria, mydriasis, etc.).

Treatment: the following criteria were used to determine type of treatment used:

Supportive: antibiotics or antivirals only, respiratory support, airway support, physical therapy, occupational therapy, IV fluids, hospitalization, speech therapy.

Steroids: use either methylprednisolone, prednisone, dexamethasone, or ACTH.

IVIg: any formulation of immunoglobulin.

PLEX: any formulation or duration of plasmapheresis.

CSF studies: studies were deemed “abnormal” in each category based on the following criteria:

Glucose < 50 mg/dL.

Protein > 45 mg/dL.

WBC ≥ 5 mm³.

Opening pressure > 20 cm H₂O.

Cytoalbuminologic dissociation: combination of an increased protein concentration (> 45 mg/dL) and a cell count < 5 mm³.

Time to recovery: all recovery times are estimated from the onset of BBE symptoms, not hospital days.

- 1st improvement: not fever by itself, extubation or isolated improvement in altered mental status only if no new treatment modalities attempted afterward (otherwise first improvement in clinical triad), time to first improvement after first nadir if course is biphasic.
- Full recovery: completely asymptomatic or “near complete recovery” accepted if only 1–2 trace symptoms remain that did not result in disability (persistent symptoms noted in appendix C).

References

1. Fisher M (1956) An unusual variant of acute idiopathic polyneuritis—syndrome of ophthalmoplegia, ataxia, and areflexia. *N Engl J Med* 255(2):57–65
2. Ito M, Kuwabara S, Odaka M, Misawa S, Koga M, Hirata K, Yuki N (2008) Bickerstaff's brainstem encephalitis and Fisher syndrome form a continuous spectrum: clinical analysis of 581 cases. *J Neurol* 255(5):674–682

3. Mori M, Kuwabara S, Fukutake T, Yuki N, Hattori T (2001) Clinical features and prognosis of Miller Fisher syndrome. *Neurology* 56(8):1104–1106
4. Koga M, Kusunoki S, Kaida K, Uehara R, Nakamura Y, Kohriyama T, Kanda T (2012) Nationwide survey of patients in Japan with Bickerstaff brainstem encephalitis: epidemiological and clinical characteristics. *J Neurol Neurosurg Psychiatry* 83(12):1210–1215
5. Hachohen Y, Nishimoto Y, Fukami Y, Lang B, Waters P, Lim MJ, Yuki N et al (2016) Paediatric brainstem encephalitis associated with glial and neuronal autoantibodies. *Dev Med Child Neurol* 58(8):836–841
6. Ajena D, Ferrari S, Romito S, Zagila F, Biban P, Squintani G (2013) A pediatric case of Miller Fisher syndrome with central involvement. *Neurol Sci* 34(9):1689–1690
7. Al-Din AN, Anderson M, Bickerstaff ER, Harvey I (1982) Brainstem encephalitis and the syndrome of Miller Fisher: a clinical study. *Brain* 105(Pt 3):481–495
8. Arias AP, Santoni A, Goenaga B, Martinez MJ, Gonzalez N (2015) Síndrome de Bickerstaff: caso Clínico. *Arch Argent Pediatr* 113(4):226–229
9. Barontini F, Sitá D (1983) The nosological position of Fisher's syndrome—ophthalmoplegia, ataxia and areflexia. *J Neurol* 229(1):33–44
10. Bell W, Van Allen M, Blackman J (1970) Fisher syndrome in childhood. *Dev Med Child Neurol* 12(6):758–766
11. Bradshaw DY, Jones HR (2001) Pseudomeningoencephalitic presentation of pediatric Guillain Barre syndrome. *J*** Child Neurol* 16(7):505–508
12. Campos-Segovia A, Perez-Villena A, Martinez J, Sanchez CZ, Blanco-Martin AB, Aurrecochea JF, Izquierdo SM, Jimenez-Martinez J (2016) Encefalitis de tronco de Bickerstaff atípica: 2 casos pediátricos. *Neurol Argent* 8(3):181–186
13. Damasceno A, Franca MC Jr, Pimenta DS, Deus-Silva L, Nucci A, Damasceno BP (2008) Bickerstaff's encephalitis, Guillain-Barré syndrome and idiopathic intracranial hypertension: are they related conditions? *Arg Neuropsiquiatr* 66(3B):744–746
14. Eggenberger ER, Coker S, Menezes M (1993) Pediatric Miller Fisher syndrome requiring intubation: a case report. *Clin Pediatr (Phila)* 32(6):372–375
15. Fargas A, Roig M, Vazquez E, Fitó A (1998) Brainstem involvement in a child with ophthalmoplegia, ataxia, areflexia syndrome. *Pediatr Neurol* 18(1):73–75
16. Fargas i Busquets A, Roig Quilis M, Gratacòs Vinyola M, Macaya Ruiz A, del Toro Riera M, Fitó Costa A (1998) Ophthalmoplegia-ataxia-areflexia in pediatrics three new patients and review of the literature. *An Esp Pediatr* 48(5):483–488
17. Fox RJ, Kasner SE, Galetta SL, Chalela JA (2000) Treatment of Bickerstaff's brainstem encephalitis with immune globulin. *J Neurol Sci* 178(2):88–90
18. Genizi J, Kasis I, Schif A, Shahar E (2007) Effect of high-dose methyl-prednisolone on brainstem encephalopathy and basal ganglia impairment complicating cat scratch disease. *Brain Dev* 29(6):377–379
19. Gologorsky RC, Barakos JA, Sahebkar F (2013) Rhomb- and Bickerstaff encephalitis: two clinical phenotypes? *Pediatr Neurol* 48(3):244–248
20. Han C, Wang Y, Jia J, Ji X, Fredrickson V, Ding Y, Sun W et al (2012) Bickerstaff's brainstem encephalitis, Miller–Fisher syndrome and Guillain-Barre syndrome overlap in an asthma patient with negative anti-ganglioside antibodies. *BMC Res Notes* 5:295
21. Kanzaki A, Yabuki S, Yuki N (1995) Bickerstaff's brainstem encephalitis associated with cytomegalovirus infection. *J Neurol Neurosurg Psychiatry* 58(2):260–261
22. Kikuchi M, Tagawa Y, Iwamoto H, Hoshino H, Yuki N (1997) Bickerstaff's brainstem encephalitis associated with IgG anti-GQ1b antibody subsequent to Mycoplasma pneumoniae infection: favorable response to immunoadsorption therapy. *J Child Neurol* 12(6):403–405
23. Kimoto K, Koga M, Odaka M, Hirata K, Takahashi M, Li J, Gilbert M, Yuki N (2006) Relationship of bacterial strains to clinical syndromes of Campylobacter-associated neuropathies. *Neurology* 67(10):1837–1843
24. Matsumoto H, Kobayashi O, Tamura K, Ohkawa T, Sekine I (2002) Miller Fisher syndrome with transient coma: comparison with Bickerstaff brainstem encephalitis. *Brain Dev* 24(2):98–101
25. Matsuo M, Odaka M, Koga M, Tsuchiya K, Hamasaki Y, Yuki N (2004) Bickerstaff's brainstem encephalitis associated with IgM antibodies to GM1b and GalNAc-GD1a. *J Neurol Sci* 217(2):225–228
26. Meyer Sauter PM, Rely C, Hackenberg A, Stahr N, Berger C, Bloemberg GV, Jacobs E et al (2014) Mycoplasma pneumoniae intrathecal antibody responses in Bickerstaff brainstem encephalitis. *Neuropediatrics* 45(1):61–63
27. Park JY, Ko KO, Lim JW, Cheon EJ, Yoon JM, Kim HJ (2014) A pediatric case of Bickerstaff's brainstem encephalitis. *Korean J Pediatr* 57(12):542–545
28. Pavone P, Le Pira A, Greco F, Vitaliti G, Smilari PL, Parano E, Falsaperla R (2014) Bickerstaff's brainstem encephalitis (BBE) in childhood: rapid resolution after intravenous immunoglobulins treatment. *Eur Rev Med Pharmacol Sci* 18(17):2496–2499
29. Ragosta K (1993) Miller Fisher syndrome, a brainstem encephalitis, mimics brain death. *Clin Pediatr (Phila)* 32(11):685–687
30. Reyes AR, Garcés P, Reig R, Martín E, Sebastián I, Herrera M (2001) Benign encephalitis of the brainstem. *Rev Neurol* 32(4):341–344
31. Rho YI (2014) Overlapping Guillain-Barré syndrome and Bickerstaff's brainstem encephalitis associated with Epstein Barr virus. *Korean J Pediatr* 57(10):457–460
32. Schain RJ, Wilson G (1971) Brainstem encephalitis with radiographic evidence of medullary enlargement. *Neurology* 21(5):537–539
33. Sonmez FM, Odemis E, Ahmetoglu A, Ayvaz A (2004) Brainstem encephalitis and acute disseminated encephalomyelitis following mumps. *Pediatr Neurol* 30(2):132–134
34. Steer AC, Starr M, Kornberg AJ (2006) Bickerstaff brainstem encephalitis associated with Mycoplasma pneumoniae infection. *J Child Neurol* 21(6):533–534
35. Suzuki K, Meguro K, Nakayama J, Aoki T, Tsurushima H (1997) MRI of an infant with Fisher syndrome. *Childs Nerv Syst* 13(2):95–96
36. Tachi N, Tsuzuki Y, Minami R (1986) A case of brainstem encephalitis with CT scan abnormality mimicking Fisher syndrome. *Brain Dev* 8(5):538–541
37. Tsapis M, Laugel V, Koob M, de Saint Martin A, Fischbach M (2010) A pediatric case of Fisher–Bickerstaff spectrum. *Pediatr Neurol* 42(2):147–150
38. Uribe SMR, Suárez HF, Sandoval RP, Mellado TP (2013) Anti-GQ1b syndrome: a report of four cases. *Rev Med Chil* 141(9):1211–1215
39. Venkateshwara Prasad KN, Venkatesh KS, Devi NG (2013) Bickerstaff brainstem encephalitis in pediatrics—a case report. *J Pediatr Neurosci* 8(3):263–264
40. Villalobos Chaves F, Casado Chocán JL, Moreno Rojas A, Serrano Castro V (1995) Miller–Fisher syndrome and encephalitis of the brainstem. *Neurologia* 10(2):115–116
41. Wali GM (1991) Bickerstaff's brainstem encephalitis associated with typhoid fever. *Postgrad Med J* 67(793):1011–1012
42. Yuki N, Odaka M, Hirata K (2000) Bickerstaff's brainstem encephalitis subsequent to Campylobacter jejuni enteritis. *J Neurol Neurosurg Psychiatry* 68(5):680–681

43. Odaka M, Yuki N, Hirata K (2001) Anti-GQ1b IgG antibody syndrome: clinical and immunological range. *J Neurol Neurosurg Psychiatry* 70(1):50–55
44. Tan IL, Mowry EM, Steele SU, Pardo CA, McArthur J, Nath A, Venkatesan A (2013) Brainstem encephalitis: etiologies, treatment and predictors of outcome. *J Neurol* 260(9):2312–2319
45. Overell JR, Hsieh ST, Odaka M, Yuki N, Willison HJ (2007) Treatment for Fisher syndrome, Bickerstaff's brain-stem encephalitis and related disorders. *Cochrane Database of Systematic Reviews* (Art. no.: CD004761)